

Predicting and Pricing Methane: Uncertainty-Aware Machine Learning for Carbon Cost Estimation in Bio-Slurry Fertilization

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Abstract

This summary presents a completed applied-ML study whose primary contribution is methodological: a small experimental dataset ($n \approx 350$, one site, one season) is used as a stress test for a validation protocol designed to make small- n predictive claims defensible. A four-learner gradient-boosting ensemble (CatBoost, XGBoost, LightGBM, Random Forest, stacked via a Ridge meta-learner) is benchmarked against a domain-standard emission-factor baseline (IPCC Tier-1) and a linear model, using nested, group-aware cross-validation to prevent temporal and replicate leakage. The primary model (CatBoost) reaches $R^2 = 0.881$ versus $R^2 = -0.174$ for the Tier-1 baseline. Beyond headline accuracy, the study runs eight targeted robustness checks — leave-one-group-out validation, learning curves, ablations, dual feature-importance methods, cross-fold stability analysis, and a complexity–interpretability trade-off comparison — each designed to attack a specific, named failure mode a reviewer would raise against a small dataset. The result is a template for interrogating one's own model rather than a claim that the model is beyond doubt.

1. Problem Framing and Data

Methane (CH_4) emitted from soil after bio-slurry fertiliser application is difficult to predict from first principles: static emission factors (the IPCC Tier-1 approach) ignore how emission rate evolves over the growing season and across application intensities. The dataset used here spans two crops (maize, napier grass), five treatments (0–100% bio-slurry substitution for mineral fertiliser, T1–T5), three field replications, and twelve weekly measurement points — 360 raw observations, reducing to approximately 350 after outlier removal. This is a genuinely small, high-noise experimental dataset, which is the point: it is treated in this study as the adversarial case a validation methodology needs to survive, not as a limitation to be minimised in the writing.

Feature engineering combines treatment/timing covariates with autoregressive terms (first- and second-order emission lags, a 3-point rolling mean, and a momentum ratio built strictly from lagged values to avoid target leakage) and non-linear time transforms. Group-aware splitting (GroupKFold on crop $\times$ treatment $\times$ replication) is used throughout so that no fold shares a replicate's time series across train and test — a leakage class that is easy to miss and that silently inflates R^2 in temporal agricultural data.

2. Model Architecture and Primary Results

Four gradient-boosting families (CatBoost, XGBoost, LightGBM, Random Forest) were each tuned via Optuna Bayesian optimisation (TPE sampler, 80 trials/model) on a held-out 20% split kept separate from the final evaluation folds. A stacking ensemble (four base learners, Ridge meta-learner, selected over Lasso/ElasticNet/ExtraTrees/Voting on held-out RMSE) was built as a robustness check, not as the headline model — CatBoost is reported as primary throughout, with stacking used only to confirm the single-model result is not an artefact of the specific learner chosen.

Model	R^2 (mean $\pm$ 95% CI)	RMSE (mg/m ² /hr)	MAPE (%)
CatBoost (primary)	0.881 ± 0.043	0.172	31.4
Stacking (Ridge meta)	0.861 ± 0.028	0.183	31.0
XGBoost	0.826 ± 0.052	0.209	35.5
LightGBM	0.783 ± 0.062	0.230	38.9
Random Forest	0.774 ± 0.078	0.239	45.9
Linear Regression	0.335 ± 0.507	0.379	40.9
IPCC Tier-1 (no ML)	-0.174 ± 0.328	0.535	117.9

Table 1 — 5-fold GroupKFold cross-validation performance, primary model vs. baselines and alternatives.

The gap between the boosted models and the linear/Tier-1 baselines is the headline result, but the more informative number is the width of the Tier-1 confidence interval (± 0.328 on a negative mean R^2): the static emission-factor approach is not merely less accurate, it is not predictive at all on this dataset, which is the standard the ML pipeline is being compared against.

3. Robustness Protocol — Eight Checks Against Small-n Failure Modes

A single cross-validated R^2 is not sufficient evidence for a 350-sample study. Each of the following checks was designed against a specific reviewer objection, rather than as generic due diligence.

Robustness check	Design	Result
Leave-One-Treatment-Out	Train on 4 treatments, test on the 5th, all 5 held out in turn	Mean $R^2=0.871$ (range 0.846–0.944) vs 0.881 in-distribution — 1.0 pt gap
Learning curve	CatBoost retrained at 20/40/60/80/100% of training data	R^2 rises 0.401→0.881 monotonically; no plateau or instability at full-n
Historical-CH₄ / agronomic-only ablation	Model refit on autoregressive lags alone (A6), then agronomic covariates alone (A7)	Both collapse to $R^2\approx 0.12$ — neither feature family alone drives the result
Permutation vs SHAP	Independent importance rankings on CatBoost (permutation) and XGBoost (SHAP)	Spearman $\rho=0.507$ rank agreement — top-3 predictors identical across methods
SHAP fold stability	Mean SHAP and coefficient of variation for each feature across 5 folds	Top 8 features CV<0.30 (stable); 6 low-importance features flagged unstable
Parsimony (CatBoost vs Stacking)	Single-model CatBoost (≈ 420 trees) vs 4-model stacked ensemble ($\approx 1,700$ trees)	CatBoost matches stacking R^2 (0.881 vs 0.864) at $\sim 4\times$ lower complexity

Table 2 — Robustness checks, each targeting a named failure mode.

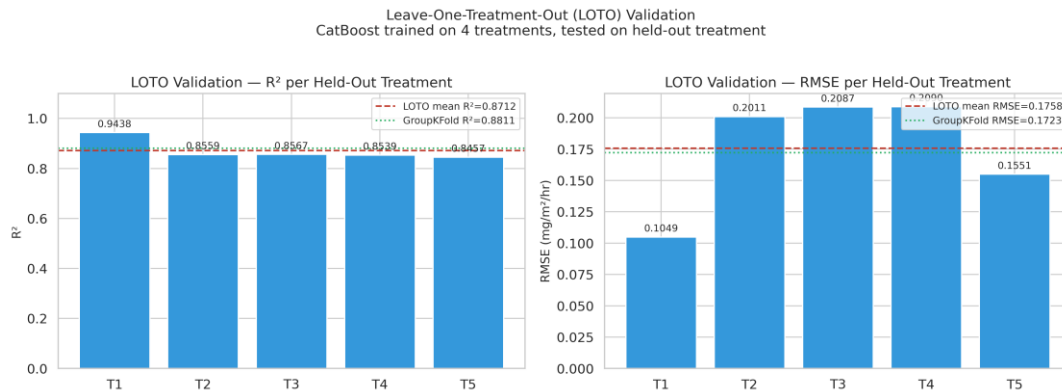


Figure 1 — Leave-One-Treatment-Out validation: R^2 and RMSE per held-out treatment, CatBoost trained on the remaining four. The 1.0-point gap to in-distribution GroupKFold R^2 indicates limited but non-catastrophic degradation under distribution shift across treatments.

4. Interpretability and Feature Stability

Feature attribution was computed two independent ways — permutation importance on the primary CatBoost model, and cross-fold-averaged SHAP values on the XGBoost base learner — to check that the model's explanation is a property of the signal rather than a specific algorithm's inductive bias. The two methods agree on the top three predictors (Emission_Roll3, Emission_Momentum, Emission_Lag2) and show moderate overall rank agreement (Spearman $\rho=0.507$), which is interpreted honestly in the write-up as moderate rather than overstated as strong.

SHAP Cross-Fold Stability Analysis (MOD 3)

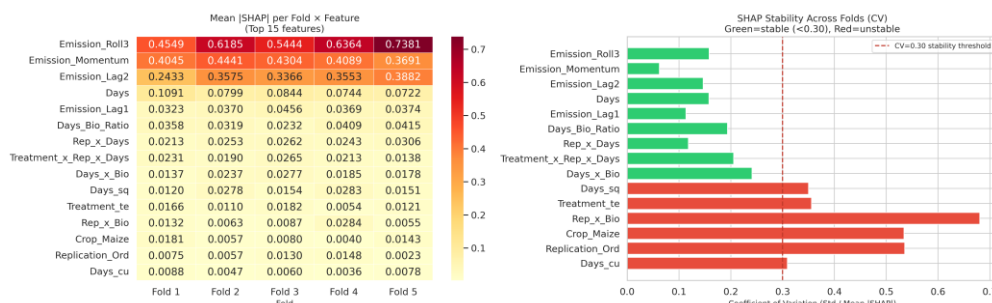


Figure 2 — SHAP cross-fold stability. The eight highest-importance features hold coefficient of variation below the 0.30 stability threshold across all five folds; six low-importance features exceed it and are flagged as unstable rather than silently included in the interpretability narrative.

5. Translating Prediction to a Decision-Relevant Quantity

As a demonstration that the pipeline output is usable beyond the ML evaluation itself, predictions were propagated (bootstrap resampling, $N=1,000$) into a social-cost-of-methane estimate per treatment under both EPA central (\$1.50/kg) and upper-bound (\$2.70/kg) valuations — an explicit, uncertainty-quantified bridge from model output to an economically interpretable number, rather than a bare point estimate presented as fact.

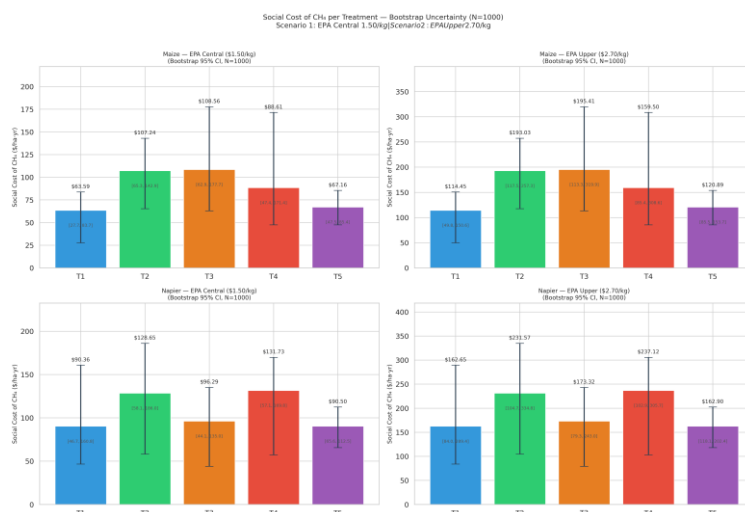


Figure 3 — Social cost of CH_4 per treatment with bootstrap 95% CI ($N=1,000$), maize and napier under both EPA scenarios. Interval widths are reported and discussed, not omitted.

6. Current Status

The experimental pipeline, all eight robustness analyses, and figure generation are complete. The manuscript is now in preparation for Computers & Electronics in Agriculture (Q1, IF 10.3) and Journal of Cleaner Production (Q1, IF 10.7).